4.2 Adding, Subtracting, and Multiplying Polynomials

Name _____ Period ____ Date _____

Algebra II 4.2 · Student sheet

I can…

- add and subtract polynomials.
- multiply polynomials, including three binomials.
- use special product patterns and Pascal's Triangle.

Part 1 · Pascal's Triangle

Build Pascal's Triangle through row 6 (each number is the sum of the two above it).

Two patterns you notice: _____

$(x + 1)^2$ = _____ $(x + 1)^3$ = _____

Part 2 · Notes: fill in the blanks

To subtract, distribute the minus sign to _____ term.

$(a + b)^3$ = _____

Row n of Pascal's Triangle gives the coefficients of _____.

Part 3 · Examples (work them with the class)

Example 1

Find the difference: $(3x^3 - 2x^2 + 5) - (x^3 + 4x^2 - x + 2)$

Example 2

Find the product: $(x - 2)(3x^2 + x - 4)$

Example 3

Find the product: $(x + 1)(x - 3)(x + 2)$

Example 4

Find the product: $(2x - 3)^3$

Example 5

Use Pascal's Triangle to expand $(x + 2)^4$.

Part 4 · Partner practice

Partner 1

Find the sum: $(x^2 + 3x - 1) + (4x^2 - 3x + 6)$

Partner 2

Find the product: $(x + 3)(x - 1)(x - 2)$